

Assignment - 2

(a) $|R \bowtie S| \leq R \times S = n \times m$

equality $\rightarrow$ no common attributes

(b) $|\pi_A(R)| \leq n$ (no. of tuples)

$\hookrightarrow$ equality holds if each tuple has a unique value for attribute A

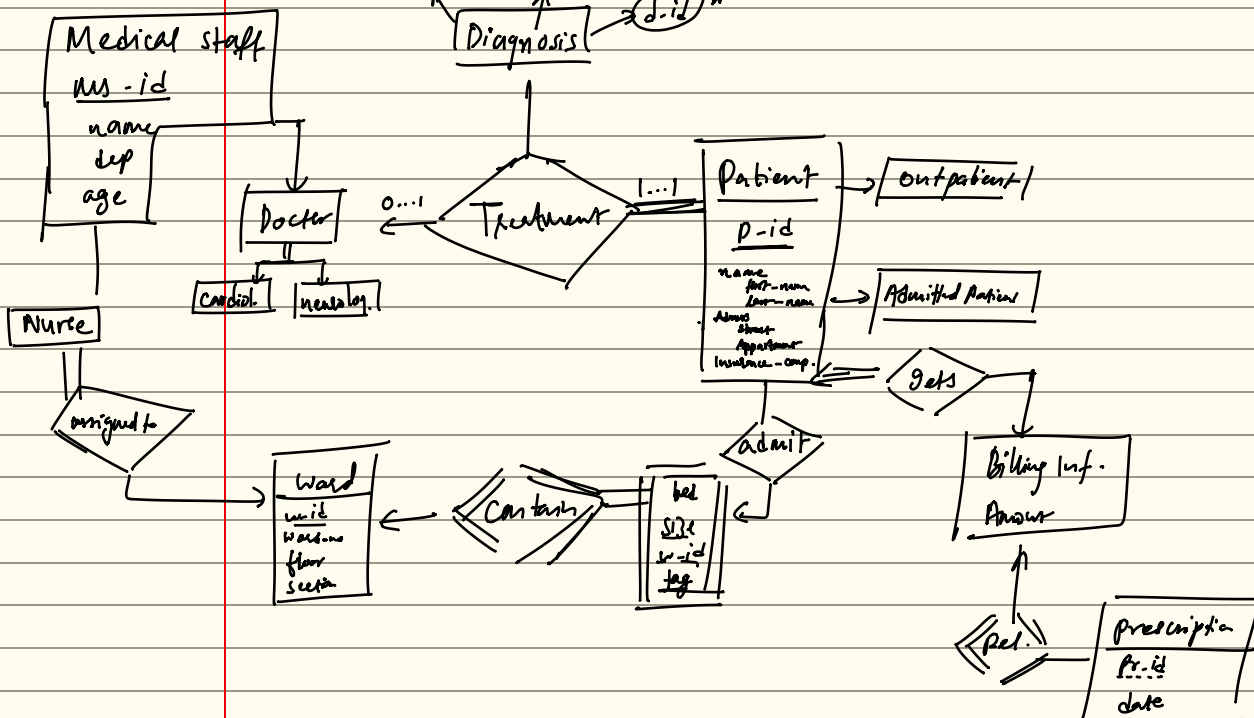
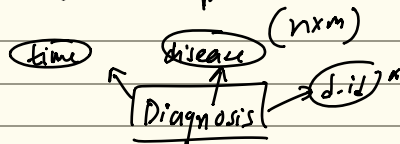
(c) $|\sigma_{A > 10}(R)| \rightarrow (\leq n)$ total.

equality when all tuples hold.

(d) $|R \cup S| \leq n + m$

equality if $|R \cap S| = 0$

(e) $R \times S \rightarrow$ cartesian product



(2) $n(U) = 1000$ $n(F) = 300$ $n(K) = 150$

$n(S) = 600$ $n(S \cap F) = 50$ $n(S \cap K) = 30$

$n(F \cap K) = 20$ $n(S \cap F \cap K) = 10$

$n(S^c \cap F^c \cap K^c) = n(U) - n(S) - n(F) - n(K)$

$+ n(S \cap F) + n(S \cap K) + n(F \cap K)$

$- n(S \cap F \cap K)$

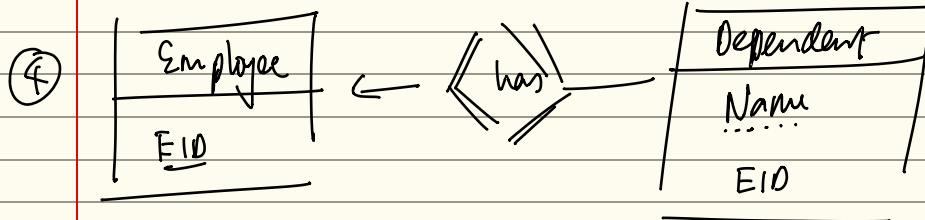
$$= 1000 - 600 - 300 - 150 + 0 + 30 + 20 + 10$$

$$= 40$$

As $40 \neq 0$

① hence $\rightarrow$ partial participation

② Overlapping as some persons exist in more than 1 specialization.



give any employee 1 dependent, then give any 1 emp! $\rightarrow$ sensory
 20dup. 21

⑤ (a) Super key : A set of attributes which can uniquely identify each tuple of relation.

instructor (ID, name, age, dept)

superkey $\rightarrow$ { ID, name } | { ID, name, age } ...

⑥ Foreign key : Attribute in child relation that references that primary key of parent relⁿ to est. relation b/w 2 tuples

⑦ Candidate key : Minimal super key

⑧ Composite key : key made by combining 2 or more attributes
 In weak

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DBMS Ass-2

6 @ $\sigma_{(O_1 \wedge O_2)} (R \times S)$

Contains tuples from $R \times S \rightarrow$ related by common attribute satisfy ^{both} O_1 & O_2 condⁿ.

$\rightarrow$ This is valid iff O_1 & O_2 are condⁿ that are indep. defined for tuples of $R \times S$.

else no.

7 @ $\{a, b\}$ (b) $\pi_{R-S}(r) - \pi_{R-S}(\pi_{P-S}(r) \times s) - \pi_{R-S}(r)$

8 (i) NULL value can be used as placeholders when value is unknown. lets assume a student table has grade written but student ~~doesn't~~ given exam. then NULL till reexan

(ii) Used to denote disjunct of values

9 @ $\pi_{personname} (\sigma_{city = "Noida"} (employee))$

(b) $\pi_{personname} (\sigma_{sal. > 7000} (works))$

(c) $\pi_{personname} (\sigma_{city = "Noida" \wedge sal. > 7000} (employee \times works))$

